

FDS6375 PMOS Circuit Analysis

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Abstract— This report analyzes and simulates a common-source (CS) MOS amplifier. The integrated circuit from Homework 1 is first examined using the Open-Circuit Time Constant (OCTC) method to determine its dominant pole and bandwidth, verified through transient and AC simulations. The assigned discrete CS amplifier is then analyzed symbolically, transistor parameters are extracted from LTspice, and component values are designed to meet the required DC conditions. Finally, analytical and simulated results are compared, and the bandwidth difference between the integrated and discrete implementations is discussed.

I. INTRODUCTION

Common-source MOS amplifiers are key analog building blocks whose performance depends on biasing and parasitic effects. This work analyzes both the integrated circuit from Homework 1 and an assigned discrete CS amplifier. The OCTC method is applied to evaluate the integrated circuit bandwidth and validate it through simulation. The discrete amplifier is then designed using extracted transistor parameters to satisfy the required DC specifications. Analytical and simulated results are compared, and the bandwidth differences are explained.

II. SYMBOLIC HIGH-FREQUENCY ANALYSIS (OCTC METHOD)

To determine the high-frequency behavior of the integrated common-source amplifier designed in Homework 1, the Open-Circuit Time Constant (OCTC) method is applied. This method approximates the dominant high-frequency pole by summing the time constants associated with each small-signal capacitance in the circuit.

The equivalent time constant is defined as:

$$\tau_{eq} = \sum_k R_k C_k \quad (1)$$

where R_k is the resistance seen by the k th capacitor when all other capacitors are open-circuited. The upper cutoff frequency is then approximated by:

$$f_H \approx \frac{1}{(2\pi \tau_{eq})} \quad (2)$$

A. Identification of Small-Signal Capacitances

For the CS PMOS amplifier with NMOS active load, the intrinsic device capacitances include:

$$C_{gsp}, C_{gdp}, C_{abp} \text{ (PMOS)}$$

$$C_{gsn}, C_{gdn}, C_{dbn} \text{ (NMOS)}$$

Among these, the output node capacitances and the gate-drain capacitance C_{gdp} are typically dominant due to the Miller effect.

B. Output Node Time Constant

The small-signal resistance seen at the output node is

$$R_{out} = r_{op} \parallel r_{on} \quad (3)$$

where r_{op} and r_{on} are the output resistances of the PMOS and NMOS transistors, respectively.

The total capacitance connected to the output node is

$$C_{out} = C_{abp} + C_{dbn} + C_{gdn} \quad (4)$$

Thus, the output time constant is $\tau_{out} = (r_{op} \parallel r_{on})C_{out}$ (5)

C. Midband Gain and Miller Effect

The midband voltage gain of the common-source stage is

$$A_v = -g_m(r_{op} \parallel r_{on}) \quad (6)$$

The gate-drain capacitance of the PMOS experiences Miller multiplication. The equivalent input capacitance due to the Miller effect is $C_{inM} = C_{gdp}(1 - A_v)$ (7)

Since the amplifier is inverting ($A_v < 0$), this becomes approximately $C_{inM} \approx C_{gdp}(1 + |A_v|)$ (8)

D. Input Node Time Constant

The total effective input capacitance is therefore

$$C_{in} = C_{gsp} + C_{gdp}(1 + |A_v|) \quad (9)$$

The resistance seen at the input is the source resistance R_{sig}

The input time constant is $\tau_{in} = R_{sig}C_{in}$ (10)

E. Total Equivalent Time Constant

The total equivalent time constant is obtained by summing the dominant contributions:

$$\tau_{eq} = \tau_{in} + \tau_{out} \quad (11)$$

Substituting the derived expressions,

$$\tau_{eq} = R_{sig}[C_{gsp} + C_{gdp}(1 + |A_v|)] + (r_{op} \parallel r_{on})C_{out} \quad (12)$$

F. Final Expression for the High-Frequency-Cutoff

The dominant high-frequency pole of the amplifier is therefore approximated by

$$f_H \approx \frac{1}{(2\pi \{R_{sig}[C_{gsp} + C_{gdp}(1 + |A_v|)] + (r_{op} \parallel r_{on})C_{out}\})} \quad (13)$$

III. BANDWIDTH FROM SIMULATION (HW1 CIRCUIT)

A. Transient Analysis

A transient frequency sweep was performed by stepping the input frequency and measuring the peak-to-peak input and output voltages. The voltage gain is computed as

$$A_v = \frac{V_{out,pp}}{V_{in,pp}} \quad (14)$$

The midband gain (low-frequency region) was found to be:

$$A_{v,mid} \approx 12.64 \text{ dB}$$

The -3 dB gain level is:

$$A_{v,mid-3} = A_{v,mid} - 3 \approx 9.64 \text{ dB} \quad (15)$$

From the transient sweep table, the gain reaches approximately 9.64 dB at: $f_{H,tran} \approx 200 \text{ kHz}$

Therefore, the bandwidth from transient simulation is:

$$f_{H,tran} \approx 200 \text{ kHz}$$

B. AC Analysis

An AC sweep was performed. The midband gain at 1 kHz was: $A_{v,mid} \approx 29.29 \text{ dB}$

The -3 dB gain level is:

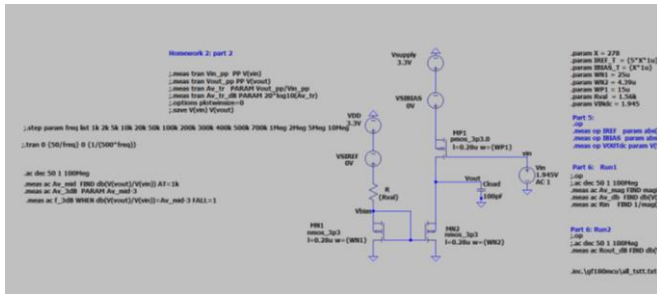
$$A_{v,mid-3} = 29.29 - 3 \approx 28.94 \text{ dB} \quad (16)$$

Using the measured -3 dB crossing, the cutoff frequency was found to be:

$$f_{H,AC} = 2.10 \times 10^5 \text{ Hz} \approx 210 \text{ kHz}$$

Thus, the bandwidth from AC simulation is:

$$f_{H,AC} \approx 210 \text{ kHz}$$



C. Comparison

The percentage difference between the two bandwidth values is:

$$\% \text{ difference} = \frac{|(210\text{k} - 200\text{k})|}{210\text{k}} \times 100 \approx 4.8 \quad (17)$$

The transient and AC methods agree closely (within approximately 5%), confirming consistency of the simulation bandwidth extraction.

IV. SYMBOLIC ANALYSIS OF THE ASSIGNED DISCRETE CS PMOS CIRCUIT

For the assigned **CS PMOS** discrete circuit (with R_S at the source to V_{SUPPLY} and R_D at the drain to ground), the following symbolic results are obtained.

A. Bias Current I_{BIAS}

By Ohm's law (same DC current through R_S and R_D):

$$I_{BIAS} = \frac{V_{SUPPLY} - V_S}{R_S} = \frac{V_{OUT}}{R_D} \quad (18)$$

In saturation (PMOS model):

$$I_{BIAS} = \frac{1}{2} k_p (V_{SG} - |V_{THp}|)^2 (1 + \lambda V_{SD}) \quad (19)$$

With:

$$V_{SG} = V_S - V_{IN}, V_{SD} = V_S - V_{OUT} \quad (20)$$

B. Small Signal Gain

For small-signal analysis, the PMOS transistor is replaced by its small-signal model, consisting of:

Transconductance g_m and output resistance r_o

The small-signal drain resistance seen at the output node is:

$$R'_D = R_D \parallel r_o \quad (21)$$

Because the source is not AC-grounded (due to R_S), source degeneration reduces the gain.

Using the small-signal model and applying KCL at the source node, the voltage gain becomes:

$$\frac{v_{out}}{v_{in}} = \frac{-g_m R'_D}{1 + g_m R_S} \quad (22)$$

The negative sign indicates that the circuit is inverting.

If channel-length modulation is included explicitly through r_o , a slightly more accurate expression is:

$$\frac{v_{out}}{v_{in}} = \frac{-g_m R'_D}{1 + g_m R_S + \frac{R_S}{r_o}} \quad (23)$$

C. Input Resistance

The MOS gate draws (ideally) zero small-signal current:

$$r_{in} \approx \infty \quad (24)$$

D. Output Resistance

With $v_{in} = 0V$ (gate at AC ground), the small-signal resistance looking into the drain is:

$$r_{out} \approx R_D \parallel (r_o (1 + g_m R_S)) \quad (25)$$

E. Bandwidth Using OCTC

Using OCTC:

$$\tau_{eq} = \sum_k R_k C_k, f_H \approx \frac{1}{2\pi \tau_{eq}} \quad (26)$$

Use the key node resistances (capacitors opened except the one under test):

$$R_G = R_{sig} \quad (27)$$

$$R_S^{eq} \approx R_S \parallel \left(\frac{1}{g_m} \right) \quad (28)$$

$$R_D^{eq} \approx R_D \parallel (r_o(1 + g_m R_S)) \quad (29)$$

Compact OCTC sum (dominant intrinsic caps):

$$\tau_{eq} \approx C_{(gs)(R_G+R_S^{eq})} + C_{(gd)(R_G+R_D^{eq})} + C_{(db)(R_D^{eq})} \quad (30)$$

Therefore:

$$f_H \approx \frac{1}{2\pi [C_{(gs)(R_G+R_S^{eq})} + C_{(gd)(R_G+R_D^{eq})} + C_{(db)(R_D^{eq})}]} \quad (31)$$

V. TRANSISTOR PARAMETERS EXTRACTION

According to the assignment, N is the sum of the digits in the ID, and the **Nth** transistor is selected after sorting the LTspice list by **Part No.** For ID **202505178**, N=30, so the selected device is the **30th** part in the sorted list: **FDS6375**.

A. Threshold Voltage V_{Th}

A DC sweep of the gate voltage was performed:

.dc VG 5 0 -0.001

The threshold voltage was measured using:

.meas dc Vth FIND V(G,S) WHEN abs(I(Vsen)) = 49u
CROSS=1

From the graph: $V_{SG,Th} = 3.48V$

Since this is a PMOS device: $V_{Th} = -3.48V$

B. Channel-Length Modulation Parameter λ

Two points were taken from the I_D vs V_{SD} curve in saturation:

Point	I_D (A)	V_{SD} (V)
A	-285.5249	4.5
B	-311.1094	5

$$\text{Average current: } I_{avg} = \frac{I_A + I_B}{2} \quad (32)$$

$$I_{avg} = -298.3172 \text{ A}$$

Channel-length modulation:

$$\lambda = \frac{I_B - I_A}{I_{avg}(V_B - V_A)} \quad (33)$$

Substituting the values: $\lambda = 0.1715 \text{ V}^{-1}$

C. Effective Gain Parameter k

Using the saturation equation:

$$I_D = \left(\frac{k}{2} \right) (V_{SG} - |V_{th}|)^2 (1 + \lambda V_{SD}) \quad (34)$$

Solving for k :

$$k = \frac{2I_D}{(V_{ov})^2 (1 + \lambda V_{SD})}$$

From the operating-point values in the simulation:

$$k \approx 6.9 \times 10^3 \text{ A/V}^2$$

D. Process Transconductance Parameter k'

From the LTspice VDMOS model of **FDS6375**, the model parameter: $Kp = 52 \text{ A/V}^2$

VI. HAND DESIGN OF R_D, R_S, V_{in} USING EXTRACTED PARAMETERS

A. Drain Resistor R_D

At DC, the drain current flows through R_D , hence:

$$R_D = \frac{V_{OUT}}{I_D} \quad (35)$$

$$R_D = \frac{2.5}{278 \times 10^{-6}} \approx 9k\Omega$$

B. Source Resistor R_S

To keep headroom and remain in saturation, choose a reasonable DC source voltage near the supply. A common choice is: $V_S = 4.8V$ (about 0.2V drop across R_S)
Then:

$$R_S = \frac{V_{DD} - V_S}{I_D} \quad (36)$$

$$R_S \approx 720\Omega$$

C. DC Input V_{IN}

Using the PMOS saturation equation (magnitudes) including channel-length modulation:

$$I_D = \left(\frac{k}{2} \right) (V_{SG} - |V_{th}|)^2 (1 + \lambda V_{SD}) \quad (37)$$

Compute:

$$V_{SD} = V_S - V_D$$

$$V_{SD} = 4.8 - 2.5 = 2.3 \text{ V}$$

$$1 + \lambda V_{SD} = 1 + 0.1715(2.3) = 1.39445$$

Solve for V_{SG} :

$$V_{SG} = |V_{th}| + \sqrt{\left(\frac{2I_D}{k(1 + \lambda V_{SD})} \right)} \quad (38)$$

Substituting:

$$V_{SG} = 3.48 + \sqrt{\left(\frac{2(278 \times 10^{-6})}{(6.92 \times 10^3)(1.39445)} \right)}$$

$$V_{SG} = 3.48024 \text{ V}$$

Finally:

$$V_{IN} = 4.8 - 3.48024 = 1.32 \text{ V}$$

VII. SMALL SIGNAL ANALYSIS OF THE AMPLIFIER

The small-signal output resistance of the PMOS transistor arises from channel-length modulation and is given by

$$r_o = \frac{1}{\lambda I_D} \approx 21k\Omega \quad (39)$$

The small-signal transconductance represents the sensitivity of the drain current to variations in the gate-source voltage and is defined as

$$g_m = \frac{\partial I_D}{\partial V_{SG}} \quad (40)$$

Using the square-law model including channel-length modulation, $g_m = k V_{ov}(1 + \lambda V_{SD}) \approx 2.32 \text{ mS}$

A. Gain A_v

With source degeneration

$$A_v = -\frac{g_m(R_D \parallel r_o)}{1 + g_m R_S} \approx -8.5 \frac{V}{V} \quad (41)$$

B. Input Resistance r_{in}

MOS gate current is ~ 0 (ignoring leakage), so:

$$r_{in} \approx \infty$$

C. Output Resistance r_{out}

Set $v_{in} = 0$ (gate at AC ground). Output resistance looking into the drain with source resistor:

$$r_{out,M2} = r_o(1 + g_m R_S) + R_S \quad (42)$$

Then the stage output resistance:

$$r_{out} = R_D \parallel r_{out,M2} \approx R_D = 9k\Omega$$

VIII. SIMULATION USING HAND-CALCULATED VALUES AND FINAL TUNING

The hand-calculated design values from Part V were first implemented in LTspice to verify the DC operating point. A DC operating-point simulation (.op) was performed to observe the resulting bias current I_{BIAS} and output voltage V_{out} . Since the device uses a practical LTspice VDMOS model, small adjustments to the resistor values and the DC input voltage were then applied to match the required targets I_{BIAS} and $V_{out} = 2.5V$. The final values listed below are the ones that satisfy the specification. Simulation snapshots of the schematic and .op results are provided.

Parameter	Calculated Parameter	Simulated Parameter
R_D	9k Ω	9.02k Ω
R_S	720 Ω	7.8k Ω
V_{IN}	1.32V	4V

--- Operating Point ---

```

(g1):      4      voltage
(s1):      5      voltage
(m001):    2.83781 voltage
(vout):    2.49816 voltage
(v00):     0      device_current
(d00):     0.003277284 device_current
(d002):    0.003277285 device_current
g(m02):    -7.04022e-10 device_current
s(m02):    -9.003277204 device_current
(v03):    -9.003277204 device_current
(s03):     0.003277284 device_current

```

IX. SMALL-SIGNAL SIMULATION USING FINAL DESIGNED VALUES

Using the final component values obtained in Part VII, the circuit was simulated in LTspice to extract the midband gain, input resistance, output resistance, and bandwidth. The gate

source V_G was biased at 4V and assigned an AC magnitude of 1V. An AC sweep was performed using:

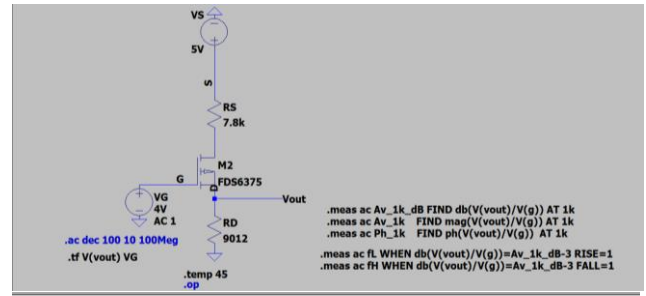
.ac dec 100 10 100Meg

The small-signal gain was evaluated as:

$$A_v(j\omega) = \frac{V(vout)}{V(g)} \quad (43)$$

At 1kHz, LTspice reported $A_v = 28.53 \text{ dB}$ with phase $\approx 150.7^\circ$, corresponding to a linear magnitude $|A_v| = 10^{\frac{28.53}{20}} = 26.7 \text{ V/V}$ at 1kHz. The input resistance was obtained from .tf yielding $r_{in} \approx 1 \times 10^{20} \Omega \approx \infty$. The output resistance was also extracted from .tf giving $r_{out} \approx 4.21 \text{ k}\Omega$. The bandwidth was determined from the -3dB point relative to the 1kHz gain, resulting in $f_L \approx 707 \text{ Hz}$ while no upper cutoff was observed up to 1GHz ($f_H > 1 \text{ GHz}$).

Small-Signal Quantity	Calculated	Simulated
Gain	$A_v \approx -8.5$	$A_v \approx -26.7$
Input resistance	$r_{in} \approx \infty$	$r_{in} \approx 1 \times 10^{20} \Omega$
Output resistance	$r_{out} \approx 9 \text{ k}\Omega$	$r_{out} \approx 4.21 \text{ k}\Omega$
Bandwidth	X	$f_L \approx 707 \text{ Hz}$



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--- Transfer Function ---

```

transfer_function:      -2.72589e-17      transfer
/g#input_impedance:     1e+20              impedance
output_impedance_at_v(vout): 4213.06      impedance

```

X. BANDWIDTH COMPARISON

The integrated circuit exhibits a much smaller bandwidth ($\approx 200 \text{ kHz}$) compared to the discrete circuit ($> 1 \text{ GHz}$). This difference is mainly due to the larger parasitic capacitances in the integrated design, which create a dominant high-frequency pole that limits the bandwidth. In contrast, the discrete circuit does not show a dominant pole within the simulated range, resulting in significantly wider bandwidth.